

Data Flow Diagram

Date	05 JULY 2026
Team ID	SWTID-2026-2423
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Data Flow Diagram

Source Component	Target Component	Data Element Passed (Arrow Label)	Description / Purpose
Student (External Entity)	Process 1: Input Processing	student query	The raw text message or code error submitted by a struggling student.
Process 1: Input Processing	Student (External Entity)	acknowledgement	Quick confirmation showing the platform received their message
Knowledge Base (Data Store)	Process 1: Input Processing	solution data	Predefined academic hints retrieved to map against the query intent.
Process 1: Input Processing	Query & Emotion Logs (Data Store)	query, sentiment tags, & user info	Storing parsed text along with detected emotion metrics (e.g., Frustrated).
Query & Emotion Logs (Data Store)	Process 2: Triage & Routing	logged student query details	Fetching raw query logs to evaluate escalation rules.
Process 2: Triage & Routing	Educator / TA (External Entity)	high-priority alert / dashboard view	Immediate notification sent to human mentors for

			high-distress cases.
Process 2: Triage & Routing	Process 3: Feedback & Engine	update status & processed records	Passing successful ticket routes to trigger appropriate response paths.
Process 3: Feedback & Engine	Student (External Entity)	confirmation & automated hint	Delivering context-aware, empathetic tips directly to the student UI.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., Student, Educator / TA).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., Analyze Sentiment & Intent, Prioritize & Route Ticket).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., Query & Emotion Logs, Knowledge Base).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed (e.g., student query, solution data).

Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.

Data Flow Diagram – Emotion Detection & Learning Support Engine

